

BJORNOYA, Svalbard and Jan Mayen

WMO: 010280

Lat: 74.504N

Lon: 19.001E

Elev: 59

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-4.5	0.3	-8.7	3.5	-4.2	-3.6	4.6	0.6	46.3	28.0	41.6	28.0	17.2	40	1.023

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	51.2	48.9	49.1	47.4	47.4	46.0	49.3	50.8	47.7	48.8	46.3	47.2	16.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
48.2	50.0	50.2	46.7	47.2	48.4	45.4	44.9	47.0	19.9	51.0	19.1	48.9	18.3	47.3	60.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.8	34.5	31.1	DB	-2.6	57.5	8.3	4.7	-8.5	60.9	-13.4	63.6	-18.0	66.3	-24.0	69.7	(4)
(5)			WB	-3.2	53.0	8.0	3.0	-8.9	55.2	-13.6	57.0	-18.1	58.7	-23.9	60.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.3	23.9	23.1	22.2	25.6	31.7	37.3	41.6	42.0	39.3	33.6	29.0	25.7
	DBStd	9.70	9.57	9.00	8.88	7.56	5.07	3.82	3.78	3.58	3.99	5.03	6.33	8.49
	HDD50	6827	810	754	861	732	566	380	261	249	320	510	630	754
	HDD65	12299	1275	1174	1326	1182	1031	830	724	713	770	975	1080	1219
	CDD50	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.7	18.5	19.1	17.7	15.9	14.0	12.1	12.7	11.9	14.7	16.3	17.4	18.4
Precipitation	PrecAvg	15.9	1.5	1.5	1.3	1.1	0.8	0.9	1.1	1.3	1.8	1.7	1.5	1.5
	PrecMax	20.8	4.9	4.4	3.1	2.2	2.0	3.6	3.3	3.4	3.7	4.1	2.9	3.7
	PrecMin	10.4	0.4	0.2	0.3	0.2	0.2	0.2	0.1	0.3	0.4	0.5	0.2	0.3
	PrecStd	2.6	0.8	0.8	0.6	0.5	0.4	0.6	0.6	0.8	0.8	0.7	0.6	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	38.0	37.6	40.3	44.1	50.3	56.3	54.2	51.7	46.4	41.4	39.2
		MCWB	37.2	36.3	35.9	38.6	41.8	46.9	52.7	51.7	49.5	44.8	39.9	37.6
	2%	DB	37.1	36.5	36.2	37.5	41.5	46.5	51.4	51.0	48.5	44.0	39.5	37.6
		MCWB	35.7	35.0	34.8	36.2	39.7	44.3	49.0	48.9	47.0	42.9	38.4	36.3
	5%	DB	35.7	35.3	35.0	36.2	39.8	44.3	48.8	49.0	46.6	42.4	37.8	36.4
		MCWB	34.5	34.0	33.8	35.0	38.1	42.3	47.1	47.6	45.5	41.4	36.8	35.1
	10%	DB	34.2	33.5	33.7	35.0	38.1	42.6	47.1	47.4	45.2	40.5	36.4	35.1
		MCWB	33.0	32.3	32.7	33.8	36.6	40.8	45.7	46.2	44.2	39.5	35.3	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	36.8	36.5	38.9	42.2	47.6	53.3	51.9	49.7	45.1	40.3	38.2
		MCDB	38.4	37.8	37.0	39.9	43.3	49.3	55.6	54.0	51.0	45.9	40.9	38.8
	2%	WB	36.0	35.4	35.1	36.6	40.1	44.6	49.3	49.4	47.3	43.2	38.7	36.6
		MCDB	36.8	36.3	35.9	37.2	41.4	46.2	51.0	50.6	48.2	44.0	39.3	37.3
	5%	WB	34.7	34.1	34.0	35.2	38.4	42.7	47.4	47.8	45.8	41.6	37.0	35.3
		MCDB	35.6	35.2	34.9	36.0	39.6	44.0	48.6	48.8	46.6	42.4	37.7	36.2
	10%	WB	33.1	32.4	32.7	34.0	36.8	41.1	45.9	46.4	44.4	39.6	35.5	34.0
		MCDB	34.1	33.4	33.7	35.0	37.8	42.4	46.9	47.3	45.1	40.4	36.3	35.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.5	7.8	7.0	5.1	4.6	5.3	4.8	4.5	4.7	5.9	7.0
		MCDBR	6.6	7.4	7.3	6.0	6.5	6.9	8.7	7.5	6.5	5.6	5.7	6.4
		MCWBR	6.7	7.6	7.3	5.7	5.8	5.9	7.1	6.4	6.0	5.7	5.7	6.6
	5% WB	MCDBR	6.8	7.2	7.6	6.0	6.5	6.6	8.6	7.5	6.4	5.7	5.7	6.1
		MCWBR	7.0	7.5	7.6	5.9	6.0	6.0	7.2	6.6	6.0	5.8	5.9	6.3
Clear-Sky Solar Irradiance	taub	N/A	0.385	0.308	0.330	0.337	0.319	0.329	0.321	0.304	0.359	N/A	N/A	
	taud	N/A	2.390	2.153	2.321	2.420	2.540	2.539	2.538	2.447	2.586	N/A	N/A	
	Ebn at Noon	0	58	186	241	260	272	263	248	202	71	0	0	
	Edh at Noon	0	7	20	26	28	25	24	21	16	6	0	0	
All-Sky Solar Radiation	RadAvg	0	23	249	690	1121	1362	1081	748	321	63	0	0	
	RadStd	0	2	28	83	89	117	96	71	35	10	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.12	+7.81	+6.17	N/A	N/A	N/A	-862	-866	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page